HIS – Claimbook Integration
Version 1.6

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OVERVIEW

AB Innovative provides Claimbook application which is used by Payers and Providers. It is a platform where the information related to PreAuth and Claims are exchanged between Providers and Payers. The related information is made available to both Payers and Providers when they have an affiliation between them.

This document explains about the HIS and Claimbook integration that will happen to seamlessly integrate the workflow.

Authentication Mechanism:

All the APIs that Claimbook will host and that HIS will host will have username and password in the header.

Alternatively, authentication token can be used

Hospital Identification:

Hospitals identification will be done by the "hopitalCode" field sent as part of the request body.
These hospital codes are mutually understood by the HIS and Claimbook.

Document Access Mechanism:

Claimbook will access the documents uploaded in HIS system via the Document service API provided by the HIS. Authentication mechanism followed in other services is expected in this api as well.
Sample service:
Api endpoint: https://his.ril.com/his/getDocumentService?_id_=12323121
Method: GET
Headers: X_Auth_Token : fsdf7sdf7ds988f7979869dsf9dsfsdsadvsavc
Response will be byte array of the file to be downloaded
IF token is invalid, should give a 403 HTTP satus.

As mentioned in the "document update api" later in this document, during the document push, for each document, the respective api service endpoint will be sent, Claimbook will invoke the api with the auth token and access the document.

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Service Hosting

HIS will host three services which will be called by Claimbook.

1. Register Case And Get Data
2. Update Claim Status
3. Get Patient Photo Upload (Details of new web service to be discussed in subsequent discussion with HIS team)

Claimbook will host three services which will be called by HIS

1. uploadDocumentInClaimbook
2. UpdatePaymentStatus
3. UpdateDischargeBill

Error Codes:

During the api communication, between the two systems, these error codes will be used to define the nature of error occurred during processing. Claimbook will expect the http status of the Rest API to be always 200 and the the error codes to be present in the response body(As mentioned in the api definitions)

Following are the codes used as of now:

200: Everything executed successfully
500: Server Error
412: Invalid request parameters, ex: invalid mrn number or Ip number or patient does not exist, etc.

More codes can be added if required and understood by both the systems.

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Below are the services hosted by HIS

1. Register Case and Get Patient Details

New Preauthorization will be initiated in Claimbook and Claimbook will invoke a service call to HIS to receive patient data.

URL	/registerCaseAndGetData
Hosted By	HIS
Header	Content-type: application/json X-User-Name: "User name" X-Pass-Phrase: "password" OR X_Auth_Token : "authentication-token"
Method	POST
Request Body	{ "hospitalCode": "AJH", "mrn": "MR-A102", "type": "IP", "caseID": "I987787688" }
Response	{ "status": 200, "hospitalCode": "RJCI", "mrn": "MR-A102", "type": "IP", "caseID": "I987787688", "patientInfo": { "patientName": "Kazi Abid Azad", "gender": "Male", "dob": "08-Mar-2018", "age": 27, "emailId": "kazi@ainnovative.com", "mobileNumber": "9600998878", "address": "BTM Layout, Bangalore", "pincode": "560029", "corporateName": "AB Innovtive Pvt. Ltd.", "employeeId": "AB002", "firNumber": "FIR-7866", "dateOfIncident": "08-Mar-2018", "insuranceCompanyName": "ICICI Lombard", "policyNumber": "PL98669", "policyType": "Self", "tpaName": "Medi Assist Pvt. Ltd.",

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```
    "tpaZone": "Bangalore",
    "tpaMemberId": "TP796966"
  },
  "medicalHistory": {
    "chiefComplaint": "This is a sample chief complaint",
    "historyOfPresentingIllness": "Sample history of
presenting illness",
    "pastHistory": "Sample past history",
    "chronicIllnessHistory": [
      {
        "name": "Diabetes",
        "duration": "2",
        "medication": "Calpol",
        "notes": "Sample notes"
      },
      {
        "name": "Hypertension",
        "duration": "4",
        "medication": "Sample medicine"",
        "notes": "Sample notes"

      }
    ],
    "personalHistory": [
      {
        "name": "Alcohol",
        "duration": "2",
        "notes": "Test smoking"
      },
      {
        "name": "Smoking",
        "duration": "4"
      }
    ],
    "familyHistory": [
      {
        "name": "Diabetes",
        "duration": "4",
        "notes": "Test Diabetes",
        "relationship": "Grand father"
      },
      {
        "name": "Hypertension",
        "duration": "4",
        "notes": "Test hyper",
        "relationship": "Mother"
      }
    ],
    "medicationHistory": [
      {
        "medication": "Test medication",
        "duration": 4,
        "durationUnit": "Year"",
        "notes": "Sample notes"

      },
```


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```
      {
        "medication": "Second Test medication",
        "duration": 2,
        "durationUnit": "Year"",
        "notes": "Sample notes"

      }
    ],
    "maternityDetails": {
      "g": 5,
      "p": 7,
      "l": 9,
      "a": 4,
      "lmp": "08-Mar-2018",
      "edd": "08-Mar-2018"",
      "notes": "Sample notes"

    }
  },
  "examinationFindingVOs": {
    "clinicalFindings": "Sample relevant clinical
findings",
    "dueToInjury": true,
    "selfInflicted": true,
    "roadAccident": false,
    "otherAccident": true,
    "substanceAbuseOrAlcoholConsumption": false",
    "notes": "Sample notes"

  },
  "provisionalDiagnosis": [
    {
      "diagnosisName": "Prv Diag 1",
      "diagnosisType": "Primary",
      "icdCode": "ICD-9669"
    },
    {
      "diagnosisName": "Prv Diag 1",
      "diagnosisType": "Primary",
      "icdCode": "ICD-9669"
    }
  ],
  "finalDiagnosis": [
    {
      "diagnosisName": "Prv Diag 1",
      "diagnosisType": "Primary",
      "icdCode": "ICD-9669"
    },
    {
      "diagnosisName": "Prv Diag 1",
      "diagnosisType": "Primary",
      "icdCode": "ICD-9669"
    }
  ],
  "billing": {
    "dateOfAdmission": "01-Mar-2018",
```


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	<pre>"dateOfDischarge": "08-Mar-2018",
	"roomType": "Delux",
	"rateCard": "GENERAL",
	"dischargeType": "On Request Discharge",
	"stayInDays": 4,
	"dateOfSurgery": "03-Mar-2018",
	"actualCostHospitalization": 1250000
	},
	"treatment": {
	"treatingDoctors": [
	{
	"department": "Cardiac Surgery",
	"doctorName": "Dr. Gaurav Shetty",
	"contactNumber": "8899876545",
	"registrationNumber": "MR878768",
	"medicalCouncil": "Some Council"
	}
	],
	"icuManagement": true,
	"dialysis": false,
	"chemotherapy": true,
	"radiation": true
	}
	}</pre>

Parameters:	hospitalCode: Unique Hospital identifier mrn : Patient MR Number type : Type of case – IP/OP caseID: IP or OP Number **Response Fields** status : 200: success, 500 failure **Fields:** dateOfDischarge – in case of IP cases roomType – in case of IP cases stayInDays – in case of IP cases treatingDoctors – At least one treating doctor should be present

2. Update Case Status

Whenever the case status changes in Claimbook application, the status will be updated to HIS system by this web-service call.

URL	/updateCaseStatusService
Hosted By	HIS
Header	Content-type: application/json X-User-Name: "User name"

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	X-Pass-Phrase: "password" OR X_Auth_Token : "authentication-token"
Method	POST

Request Body	{
	"hospitalCode": "AJH",
	"mrn": "MR-A102",
	"type": "IP",
	"caseID": "I987787688",
	"state" : "SUBMIT_ORIGINAL",
	"statusChangeTime" : "08-Mar-2018 16:32:99"
	}

Response	{
	"status": 200,
	"message" : "State updated successfully"
	}

Parameters:	hospitalCode: Unique Hospital identifier mrn : Patient MR Number type : Type of case – IP/OP caseID: IP or OP Number Below are the values of **state:** **SUBMIT_ORIGINAL:** New Preauth submission **SUBMIT_ENHANCED:** Enhancement Preauth submission **SUBMIT_DISCHARGED:** Discharged Preauth submission **SUBMIT_DISCHARGED_REVISED:** Discharged Revised Preauth submission **RESUBMIT_ORIGINAL:** Resubmitted Preauth **SUBMIT_CLAIM:** Claim Submit **Response Fields** status : 200: success, 500 failure

3. Get Patient Photo from HIS to Claimbook

URL	/getPatientPhotoService
Hosted By	HIS
Header	Content-type: application/json

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	X-User-Name: "User name" X-Pass-Phrase: "password" OR X_Auth_Token : "authentication-token"
Method	POST

Request Body	{
	"hospitalCode": "AJH",
	"mrn": "MR-A102"
	}

Response	{
	"status": 200,
	"photoDetails" : {
	"name" : "photo.jpg",
	"path" : " https://his.ril.com/his/getDocumentService?_id_=12323121"
	}
	}

Parameters:	hospitalCode: Unique Hospital identifier mrn : Patient MR Number type : Type of case – IP/OP caseID: IP or OP Number **Response Fields** status : 200: success, 500 failure

Below are the services hosted by claimbook

1. Document update from HIS to Claimbook

Whenever any document is uploaded or gets created in HIS system, it will uploaded to Claimbook server.

URL	/updateDocumentService
Hosted By	Claimbook
Header	Content-type: application/json X-User-Name: "user name" X-Pass-Phrase: "password"

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	OR X_Auth_Token : "authentication-token"
Method	POST

Request Body
 	
Message from Claimbook system to HIS system
	{
	"hospitalCode": "AJH",
	"mrn": 78768,
	"caseID": "115A",
	"type":"IP",
	"documents":[{
	"name": "photo.jpg",
	"path": "https://his.ril.com/his/getDocumentService?_id_=x798989",
	"type":"SIGNED FORM",
	"description" : "This is signed form"
	},
	{
	"name": "final-bill.pdf",
	"path": "https://his.ril.com/his/getDocumentService?_id_=xioo6776",
	"type":"FINAL BILL",
	"description" : "Final Bill"
	}
	]
	}

Response	Success:
	{
	"status": 200,
	"message" : "document upladed successfully"
	}
	Failure:
	{
	"status" : 500,
	"message" : "Reason",
	}
	OR
	{
	"status" : 412,
	"message" : "Invalid IP number",
	}

Description	This service will be called by HIS whenever a new document is generated or uploaded. The **path** defines the api endpoint where the document is available. This api must be secure with authentication token. Claimbook will call these api and store the document in Claimbook repository.

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Parameters:	hospitalId : Unique Hospital identifier mrn : Patient MR Number type :IP –In-Patient OP - Outpatient caseID: IP or OP Number **type in document** : These are the document categories shared between the two systems Following are the possible values: TPA/ESI MEMBERSHIP CARD PHOTO IDENTITY CARD SIGNED FORM DOCTORS LETTER INTERIM BILL DISCHARGE SUMMARY INVESTIGATION REPORT FINAL BILL REVISED FINAL BILL OPTIONAL DOC APPROVAL LETTER SURGICAL PHARMACY BILL PATHALOGY TEST REPORT REJECTION LETTER

2. Claim Payment update from HIS to Claimbook

When payment (patient payment or TPA claim realized payment or write off adjustment amount) is received and/or updated in HIS Billing for a insurance billing case, HIS will call Claimbook service to push the insurance bill payment details in Claimbook system. The service "updatePaymentService" will be hosted in Claimbook web service which will be invoked by HIS multiple times at the time when payment is updated in HIS Billing

URL	/updatePaymentService
Hosted By	Claimbook
Header	Content-type: application/json X-User-Name: "hospital user name" X-Pass-Phrase: "hospital password" OR X_Auth_Token : "authentication-token"
Method	POST

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Request Body
 	
Message from Claimbook system to HIS system
	{
	"hospitalCode" : "1",
	"mrn" : "696979797798",
	"type" : "IP",
	"caseID" : "101A",
	"payments" : [
	{
	"paymentSource" : "TPA",
	"paymentDate" : "21-Feb-2018",
	"transactionNumber" : "TR-99798866",
	"transactionType" : "NEFT",
	"utrNumber" : "UTR9979",
	"bank" : "AXIS",
	"amount" : 65000.00
	},
	{
	"paymentSource" : "PATIENT",
	"paymentDate" : "17-Feb-2018",
	"transactionNumber" : "BIL-8866",
	"transactionType" : "RTGS",
	"utrNumber" : "UTR9979",
	"bank" : "",
	"amount" : 5000.00
	}
	]
	}

Response	Success:
	{
	"message": "Payment information successfully",
	"status": 200
	}
	Failure:
	{
	"status" : 500,
	"message" : "Reason",
	}

Parameters:	hospitalId : Unique Hospital identifier mrn : Patient MR Number type :IP –In-Patient OP - Outpatient caseID: IP or OP Number paymentSource: Either "TPA" or "PATIENT" transactionType : NEFT, RTGS, Cheque

3. Discharge Final Bill update from HIS in Claimbook

HIS will update final Bill (detailed breakup) by invoking interface "updateBillingService" web-service which will be deployed in Claimbook web server. This web service will be invoked by HIS when final bill is generated in HIS.

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There are 2 ways in which billing items data can be passed.

1. Billing Item details in JSON format
2. Billing Item details can be as a file (Currently CVS format) along with other billing data in json

1. When billing items are sent as JSON

URL	/updateBillingService
Hosted By	Claimbook
Header	Content-type: application/json X-User-Name: "hospital user name" X-Pass-Phrase: "hospital password" OR X_Auth_Token : "authentication-token"
Method	POST
Request Body **Message from Claimbook system to HIS system**	{ "hospitalCode": "Hospital_code", "mrn": 78768, "caseID": "101A", "billingInfo": { "dateOfAdmission": "11-Jan-2018", "dateOfDischarge": "20-Jan-2018", "roomType": "General", "rateCard": "RATE-1", "dischargeType": "DAMA (Discharge against Medical Advice)", "stayInDays": 3, "dateOfSurgery": "13-Jan-2018", "actualCostHospitalization": 890000.0, "finalBillNumber" :"BL987797", "finalBillDate" : "09-Mar-2018", "finalBillAmount" : 34000.00, "billingItems": [{ "itemName": "Delivary Normal - with episiotomy / Prof. Charges (Dr. T.K Yogendra Kumar)", "itemNameFromServiceMaster": "Prof. Charges", "serviceCode": "MI 00055", "roomType": "General", "rate": 8000, "quantity": 1, "charges": 8000, "billingDate": "23-Jan-2018" },{ "itemName": "Delivary Normal - with episiotomy / Hospital Charges", "itemNameFromServiceMaster": "Hospital Charges", "serviceCode": "HC IN 902",

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	"roomType": "General", "rate": 4500, "quantity": 1, "charges": 4500, "billingDate": "7-Jan-2018" }] } }

Response	Success:
	{
	"message": "Billing Information updated successfully",
	"status": 200
	}
	Failure:
	{
	"status" : 500,
	"message" : "Reason",
	}

Parameters:	hospitalCode: Unique Hospital identifier mrn : Patient MR Number caseID: IP or OP Number **itemName**: This is the item name that gets printed on the final bill PDF file. **itemNameFromServiceMaster:**This is the corresponding name from the service master of the HIS system. **serviceCode**: This is the corresponding service code from HIS system

2. In File(CSV)

URL	/updateBillingService
Hosted By	Claimbook
Header	Content-type: application/json X-User-Name: "hospital user name" X-Pass-Phrase: "hospital password" OR X_Auth_Token : "authentication-token"
Method	POST

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Request Body
 	
Message from Claimbook system to HIS system
	{
	  "hospitalCode": "1",
	  "mrn": 78768,
	  "type": "OP",
	  "caseID": "101A",
	  "billingInfo": {
	  "dateOfAdmission": "11-Jan-2018",
	  "dateOfDischarge": "20-Jan-2018",
	  "roomType": "General",
	  "rateCard": "RATE-1",
	  "dischargeType": "DAMA (Discharge against Medical Advice)",
	  "stayInDays": 3,
	  "dateOfSurgery": "13-Jan-2018",
	  "actualCostHospitalization": 890000.0
	  },
	  "file": {
	  "name": "Final-Bill.csv",
	  "path": "http://www.his.com/final-bill-2.csv"
	  }
	}

Response	Success:
	{
	  "message": "Billing Information updated successfully",
	  "status": 200
	}
	Failure:
	{
	  "status" : 500,
	  "message" : "Reason",
	}

Parameters:	hospitalCode: Unique Hospital identifier mrn : Patient MR Number type :IP –In-Patient OP - Outpatient caseID: IP or OP Number file: this contains the Billing File details. path: downloadable file path

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